Machine Learning for Heterogeneous Agent Macroeconomics
Reading List

Duke University Mini-Course · September 8–11, 2026

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Before the Mini-Course (Suggested Pre-Reading & Pre-Coding)

The lectures will be much easier to follow — and the discussion much more fun — if you have had a look at these two items beforehand. If your time is limited, the coding item is the one that pays off most.

- **Reading — heterogeneous-agent models.** Dirk Krueger, *An Introduction to Macroeconomics with Household Heterogeneity* (lecture notes), 2026. [link]

  Chapter 6, "The SIM in General Equilibrium," is the most directly relevant; Chapter 4, "The SIM in Partial Equilibrium," is a useful background reference if the material is new to you.

- **Coding — neural networks for economic models.** Simon Scheidegger, tutorials on using neural networks to solve simple Brock–Mirman models (Jupyter notebooks).

  If you can, read and run both notebooks: Notebook 1 — deterministic Brock–Mirman, and Notebook 2 — stochastic Brock–Mirman. New to Python? See the Python refresher here.

Lecture 1 — DeepHAM: Global Solutions with Generalized Moments

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks.* Quantitative Economics 17(2), 297–341. Code: github.com/frankhan91/DeepHAM. [link]

  Neural networks approximate the value and policy functions, and the cross-sectional distribution is represented by a small set of *learned* generalized moments; the objective is optimized directly along simulated paths. Focus on the moment representation, the simulation-based training objective, and the constrained-efficiency application.

Lecture 2 — Structural Reinforcement Learning

- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics.* arXiv:2512.18892. Tutorial code: github.com/StructuralRL/SRL. [link]

  Replaces the cross-sectional distribution with low-dimensional *prices* as state variables, lets agents learn equilibrium price dynamics from simulated paths, and proposes a policy-gradient method that exploits agents' structural knowledge of their own dynamics. Sidesteps the Master equation; applications to Krusell–Smith, Huggett with aggregate shocks, Huggett with portfolio choices (two nontrivial market clearing conditions), and HANK.

Lecture 3 — DeepSAM: Search and Matching with Distributions

- Payne, Rebei & Yang (2026). *Deep Learning for Search and Matching Models.* Conditionally accepted, Econometrica. [link]

Characterizes general equilibrium of search and matching models with aggregate shocks and two-sided heterogeneity as a high-dimensional PDE with the distribution as a state variable, then solves it globally by deep learning and estimates parameters via simulated method of moments. Applications to labor search: distributional feedback and amplification, and sorting over the business cycle.

Background for the Whole Course

Heterogeneous-agent models

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving.* Quarterly Journal of Economics 109(3), 659–684. [link]

  The foundational discrete-time, incomplete-markets heterogeneous-agent model.

- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy.* Journal of Political Economy 106(5), 867–896. [link]

  The classic heterogeneous-agent model with aggregate risk. Its curse of dimensionality is exactly what Lectures 1 and 2 are designed to overcome, and it is the lead application in both.

Deep learning for economic models

- Fernández-Villaverde (2026). *Deep Learning for Solving Economic Models.* Journal of Economic Literature, forthcoming. [link]

  A broad survey of how deep learning is used to solve dynamic economic models, with code on the companion page.

- Maliar, Maliar & Winant (2021). *Deep Learning for Solving Dynamic Economic Models.* Journal of Monetary Economics 122, 76–101. Code: github.com/marcmaliar/deep-learning-euler-method-krusell-smith. [link]

  The "all-in-one expectation operator" formulation: train a neural network on simulated paths to satisfy the model's equilibrium conditions.

- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets.* International Economic Review 63(4), 1471–1525. Code: github.com/sischei/DeepEquilibriumNets. [link]

  Neural networks trained to satisfy equilibrium conditions along simulated paths, with a wide range of applications.

- Goodfellow, Bengio & Courville (2016). *Deep Learning.* MIT Press (full text free online). [link]

  The canonical deep-learning textbook.

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models.* arXiv:2406.13726 (forthcoming, Journal of Political Economy: Macroeconomics). [link]

  Introduces Economic-Model-Informed Neural Networks (EMINNs) to compute global solutions of master equations, with the continuous-time Krusell–Smith model as the lead application.

- Kargar, Passadore, Silva & Yang (2026). *Liquidity and Risk in OTC Markets: A Theory of Asset Pricing and Portfolio Flows.* Working paper. [link]

  An OTC application of the search-and-matching machinery in Lecture 3: asset pricing and portfolio flows in over-the-counter markets with liquidity and risk.

Reinforcement learning

- Moll (2026). *Lectures on Reinforcement Learning for Economists.* Lecture notes and slides. [link]

  The most direct entry point to the RL material behind Lecture 2, written for economists rather than for computer scientists.

- Zhao (2024). *Mathematical Foundations of Reinforcement Learning.* Springer Nature & Tsinghua University Press (esp. Ch. 6 on stochastic approximation / Robbins–Monro). [link]

  Excellent RL textbook.

- Wibault, Forkel, Towers, Wibault, Duque, Whittle, Schaab, Yang, Wang, Osborne, Moll & Foerster (2026). *Recurrent Structural Policy Gradient for Partially Observable Mean Field Games.* International Conference on Machine Learning (ICML), 2026. Code: github.com/CWibault/mfax. [link]

  A history-aware "structural" policy-gradient method for partially observable mean-field games with common noise, and the MFAX (JAX) framework. The CS companion paper to Lecture 2.